

Markscheme

Mathematics: analysis and approaches

Practice question bank

136. (a)

Given that $z = r(\cos \theta + i \sin \theta)$, where $r = 4$ and $\theta = \frac{2\pi}{3}$.

Find z in the form $a + bi$, giving a exactly and b correct to 3 significant figures.

$$a = 4 \cos \left(\frac{2\pi}{3} \right) = 4(-0.5) \quad (M1)$$

$$a = -2 \quad \text{A1}$$

$$b = 4 \sin \left(\frac{2\pi}{3} \right) = 4 \left(\frac{\sqrt{3}}{2} \right) \quad (M1)$$

$$b \approx 3.46 \quad \text{A1}$$

[4 marks]